

DSA 210 – Project Proposal

Investigating the Correlation Between Weather Conditions and Personal Music Listening Habits

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Motivation

This project explores whether local weather conditions influence personal music listening preferences. Music is widely regarded as a reflection of mood, and environmental factors like weather are known to affect emotional states. By merging personal Spotify data with historical weather records, I aim to discover if patterns exist between atmospheric variables (temperature, precipitation, cloud cover) and the musical characteristics (valence, energy, tempo) of the tracks I listen to.

Data Sources and Collection

Spotify Data: I will export my streaming history via Spotify's Privacy dashboard. This includes track names, artist names, and timestamps for every song played. I will also use the Spotify library to fetch audio features (valence, energy, tempo) for each track.

Weather Data (Enrichment): Historical hourly weather data will be retrieved from Visual Crossing or OpenWeatherMap for my location. Each listening event will be matched with the corresponding weather record, adding temperature (°C), precipitation (mm), humidity (%), and weather description (e.g., Sunny, Rainy).

Data Characteristics

The merged dataset will contain one row per stream event with columns spanning listening metadata (timestamp, track, artist), audio features (valence, energy, tempo), and weather variables (temperature, precipitation, humidity, condition). Expected size is approximately 800-1000 records.

Planned Methodology

I will perform EDA to visualize trends (e.g., rainy days vs. low-valence music), then conduct hypothesis tests (t-tests, Mann–Whitney U) to assess statistical significance. In the ML phase, I will apply regression to predict audio features from weather inputs and clustering (K-Means) to discover natural listening–weather groupings. All code will be in Python and maintained on GitHub.